

Find similar items paris - Project report

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Chapter 1

Introduction

The main objective of this project is to implement a scalable, data-mining-oriented, comparison algorithm to detect similar text based items. The core computation consists of finding the candidate pairs in a document collection and verifying the pair true Jaccard similarity, that, using parameters optimization techniques, is within a certain threshold bound specified by the user.

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1.1 Dataset description

The dataset to test the algorithm is a collection of New York Times articles and comments hosted on kaggle at the URL:

<https://www.kaggle.com/datasets/benjaminawd/new-york-times-articles-comments-2020>.

The specific dataset version used is the version 6, published in 2021, the tests and the report have been done during September 2026.

The dataset contains two types of objects: **Article** and **Comment**. The implemented notebook only shows an example on **Articles** abstract, but the pipeline can be easily adapted to different types of textual documents. The specific **Article** attributes used are: (`uniqueId:str`, `abstract:str`), this tuple of (`docId`, `docText`) is the interface of the implemented pipeline.

1.1.1 Data organization and assumptions

The following assumptions has been made in order to contextualize a generic **abstract** object: is assumed to be at least one sentence long, includes more than one word and

punctuation marks, the sentence are well structured and contains very few spelling mistakes. Also the assumptions about the **abstract** size are that it is sufficiently long to allow a word based processing, but small enough such that at least one whole **abstract** can be loaded in the main memory in order to be processed.

The next chapter will delve into an algorithm that, given the above assumptions, is able to scale horizontally on the number of articles and to retrieve only candidate pairs with specified Jaccard similarity threshold.

Chapter 2

Implementation

2.1 Implemented algorithm

The following section contains details about the adopted techniques, the design choices and an overview of data flow execution. To find similar text items, a similarity approximation filtering candidate pairs search pipeline was chosen and implemented from scratch. This pipeline is mainly inspired by the one described in the section 3.4.3 of the book *Mining of Massive Datasets* [1], but, in addition, such pipeline has been extended with these two main features:

- Map reduce approach: the pipeline and relative modules are designed to run on a distributed map reduce job on a PySpark cluster. In this way, assuming that the cluster can scale up horizontally, the dataset can grow as much as needed.
- Simple implementation of locality sensitive hashing parameters tuning based on bands and rows parameters optimization, and as consequence also the length of signature vector. The bands and rows are computed as a function of the desiderate threshold and error. Through this tuning, based on the specified optimization objective, the signature size is tuned to be either **precise**, minimizing the delta with the Jaccard similarity, or **compressed**, to limit the memory and reduce the computation time.

2.1.1 Pipeline and execution overview

The full implementation details of the grid search are in the section 2.1.2, as an overview we will just assume that we have found the tuple (n, b, r) , which represents, in order: the signature length, the number of bands in which the signature is divided and the number of rows in each band. The distributed execution follows this work flow:

1. The CSV file gets loaded in the spark cluster in a distributed way across the cluster.
2. The data frame is converted to a key value tuple in the form of (id, text) and a map step, using the core pipeline of custom modules, calculates the value of the LSH on the text signature. More details on custom modules and the candidate pairs matches can be found in the section 2.2.

3. Now the tuple is in the form $(id, text, list[(band, bucket)])$ and using a flatMap we can now achieve the tuple with as key the tuple $(band, bucket)$ and as value a list that contains the id.
4. Executing a simple reduceByKey all the documents with the same band and bucket are aggregated. It is important to note that until this step all the execution was fully distributed without any communication needed in the cluster.
5. In the end, we obtain the tuples in the form $((band, bucket), [...ids])$, now creating all the possible distinct pairs using the list of documents we can check the results. To avoid unnecessary lookups during verification, the implemented pipeline keeps the initial tuple $(id, text)$ across the whole pipeline not just the document id. it is not necessary but to perform results evaluation it is crucial to recalculate the Jaccard similarity of the pair and evaluate the pair.

2.1.2 Locality sensitive hashing parameter search

To optimize the LSH parameters, a set of hyper parameters has been defined as:

- K : *Shingle length*. The length of each shingle in the K -gram representation.
- T : *Shingler type*. The shingling technique for text processing partitioning: $T \in \{\text{char}, \text{word}\}$.
- B : *Hash bucket size in bytes*. The hash bucket capacity in bytes per hashed shingle value to minimize collisions.
- O : *Optimization objective*. The target metric optimization function: $O \in \{\text{precision}, \text{compression}\}$.
- t : *Similarity threshold*. The Jaccard similarity threshold $t \in [0, 1]$, where document pairs with an estimated similarity score $J \geq t$ are classified as matches.
- ϵ : *Tolerable error margin*. The target maximum allowable estimation error for the minhash similarity.
- $N_{\max}$: *Maximum signature length*. The upper bound constraint on the signature vector size $n \leq N_{\max}$, capping the maximum number of independent hash functions evaluated during minhash signature construction.

Here is a summary of the implemented function that shows the core bound and optimization techniques.

Function input and output

Input:

- $t \in (0, 1]$: target Jaccard similarity threshold
- $N_{\max} \in \mathbb{N}$: maximum signature length bound
- $O \in \{\text{precision}, \text{compression}\}$: optimization objective
- $\epsilon \in (0, 1]$: tolerable threshold error margin

Output:

- Sorted list V of valid parameter tuples:
 $(n : \text{signature size}, b : \text{bands}, r : \text{rows}, t_{\text{real}}, t_{\text{error}}, e_{\text{Jaccard}})$

Step 1: find signature lower bound $n_{\min}$

We can see the minhash signature of length n as n independent minhash functions that allows us to estimate the Jaccard similarity. Then we can estimate the error as $E_J \leq \epsilon$, so if we set:

$$E_J = \frac{1}{\sqrt{n}} \leq \epsilon \implies n \geq \frac{1}{\epsilon^2} \implies n_{\min} = \left\lceil \frac{1}{\epsilon^2} \right\rceil$$

finding a signature length lower bound, based on statistical lower bound, that ensure bounded error.

Step 2: find rows per band upper bound ($r_{\max}$)

From the threshold approximation of the LSH banding $b_{\text{ideal}} = t^{-r}$, the total estimated signature length as a function of r is:

$$\hat{n}(r) = b_{\text{ideal}}(r) \cdot r = r \cdot t^{-r}$$

Fixing $b \geq 2$, we have $r = \frac{n}{b} \leq \frac{n_{\max}}{2}$ and by incrementing r from $r = 2$ until $\hat{n}(r) > n_{\max}$, setting $r_{\max}$ as the previous valid integer cause $\hat{n}$ is strictly monotonic increasing r .

Step 3: discrete grid search in interval $r \in [2, r_{\max}]$

For each integer $r \in [2, r_{\max}]$:

$$\begin{aligned} b_{\text{ideal}} &= t^{-r} \\ b &= \text{round}(b_{\text{ideal}}) \\ n &= b \cdot r \end{aligned}$$

Filter valid solutions requiring $b \geq 2$ and $n \geq n_{\min}$, then evaluate metrics:

$$\begin{aligned} t_{\text{real}} &= b^{-1/r} \\ E_t &= |t_{\text{real}} - t| \\ E_J &= \frac{1}{\sqrt{n}} \end{aligned}$$

Step 4: sorting results based on optimization objective

Given the candidate parameter combinations on the feasibility curve, they get sorted based on the desired optimization objective.

- **Compression:** Minimizes memory footprint and hash evaluation cost by selecting the smallest valid signature length $n = b \cdot r$.
- **Precision:** Minimizes the minhash statistical error $E_J = \frac{1}{\sqrt{n}}$ by maximizing n , reducing variance in similarity estimation.

2.2 Modules description

This section outlines the core modules implemented from scratch that allows to write with a much higher level of abstraction the map reduce job.

2.2.1 Shingler Module

Configures and processes text using a character or word based sliding window to extract shingles.

- **ShinglerConfiguration**: Configures window size (k), shingling mode ('char' or 'word').
Optional pre processing (**sanitizer**), and hashing (**hasher**).
Validation rules enforce valid types, positive integers for k , and callable objects for pre processing or hashing functions.
- **Shingler**: Executes shingling operations based on a **ShinglerConfiguration**.
 - **get_shingles**(*text: str*) $\rightarrow$ *list[str]*: Sanitizes and splits text into shingles of size k .
Returns an empty list if the text is shorter than k .
 - **get_hashed_shingles**(*text: str*) $\rightarrow$ *list[int]*: Extracts shingles, applies the configured **hasher** function, and returns a list of integer hash values.
Returns an empty list if the text is too short.
Requires a configured **hasher** function.

2.2.2 Hash permutator and minhash generator

Generates linear hash functions to simulate set permutations and builds minhash signatures.

- **PolynomialHashFunction**: Encapsulate linear coefficients (a, b) and calculate bucket bound hash values via $h(x) = (a \cdot x + b) \bmod \text{mod-factor}$.
- **HashPermutator**: Manages n unique, pseudo random polynomial hash functions under a given modulo factor.
 - **hash**(*index: int*) $\rightarrow$ *list[int]*: Maps a single input index across all generated hash functions, returning a list of permuted hash values.Configurations require positive integer inputs for function counts and bounds. Errors are raised if unique seeds cannot be derived or requested limits are exceeded.
- **MinhashGenerator**: Static utility for signature creation.
 - **get_signature**(*permutator: HashPermutator, shingles: list[int]*) $\rightarrow$ *list[int]*: Computes the minhash signature vector by extracting minimum hash values across all input shingles for each permutation.

2.2.3 Locality Sensitive Hasher (LSH)

Divides minhash signatures into bands and rows to efficiently detect candidate duplicate pairs.

- **LocalitySensitiveHasher**: Divides a minhash signature into b bands of r rows to index potential matches.

- **get_estimated_threshold()** $\rightarrow$ *float*: Returns the estimated Jaccard similarity threshold, calculated as $(1/b)^{(1/r)}$.
- **hashed_bands**(*signature: list[int]*) $\rightarrow$ *list[tuple[int, int]]*: Splits the signature vector into b bands of r rows, returning a list of (**band_index**, **bucket_hash**) tuples.

Configurations enforce positive integer bounds and strictly require that `signature_size =  $b \cdot r$` . Input signatures must match the configured length and item byte capacity constraints.

Chapter 3

Experiments

The following section is a list of the key experiments that has been conducted using the implemented algorithm on the `Article` objects of the dataset, also every experiment had a limit of 2500 csv lines. For each experiment are reported: the hyper parameters configuration, to allow reproducibility of the experiment, the parameter tuning results, the results and the first 5 found pairs, if any.

3.1 Experiment 1

In this experiment we will test the pipeline using a 3-grams shingle based on chars. The desiderate threshold to 0.7 with an error of 0.1 so all solution with a Jaccard similarity ≥ 0.6 should be considered as true positive results.

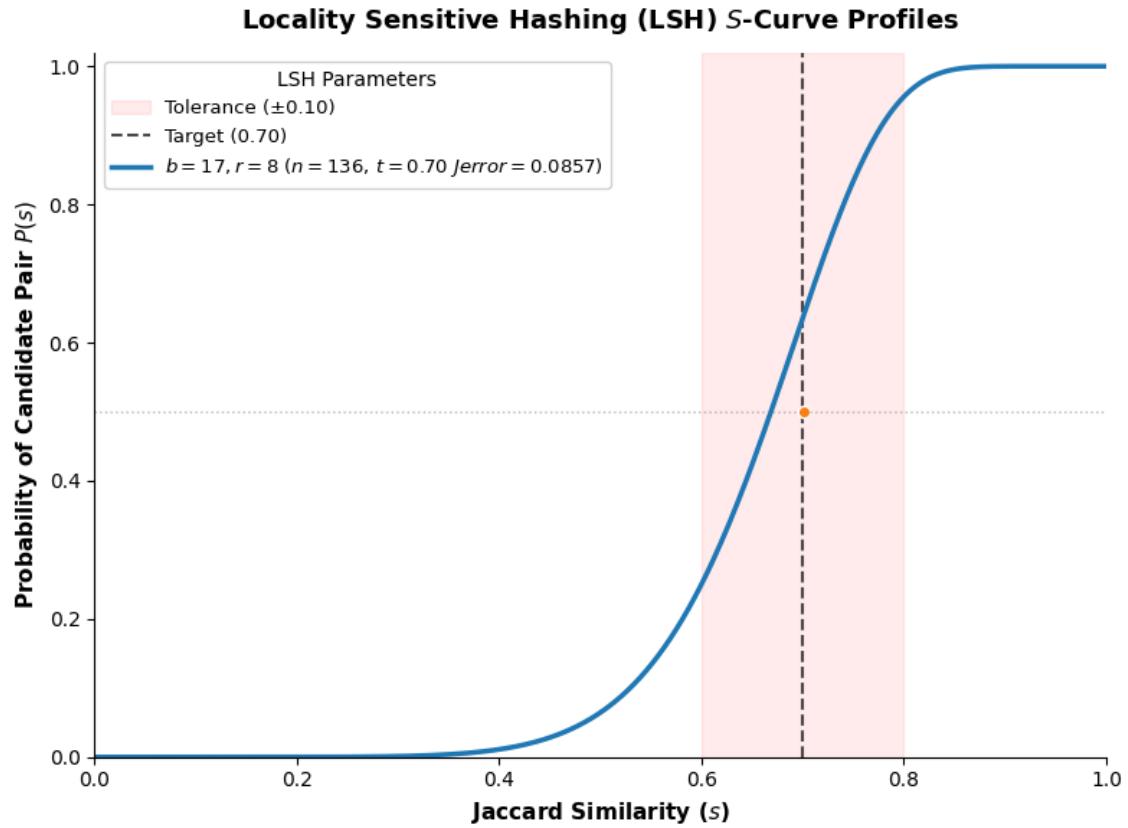
3.1.1 Hyper parameters

```
K = 3
SHINGLER_TYPE = "char"
SHINGLER_BUCKET_SIZE_IN_BYTES=4
OPTIMIZATION_OBJECTIVE = "precision"
THRESHOLD = 0.7
ERROR = 0.1
MAX_SIGNATURE_LEN = 200
```

3.1.2 LSH parameters tuned on threshold

```
CLUSTER CONTEXT:
Shingles:
  k: 3
  type: char
  hashed shingle bucket size in bytes: 4
Minhash:
  signature length: 136
LSH:
  bands: 17
```

rows: 8



3.1.3 Candidate pairs execution log

All candidate pairs found: 92

All non identical pairs found: 27

First 5 non identical pairs

Pair 0:

ID1: nyt://article/82928e0e-bc1f-5daf-8a62-027a45e486e1

ID2: nyt://article/547b64d3-7947-5205-a3a7-9dc2b7f31481

Text1: A new program lets students customize their meals,
participate in taste tests and brainstorm ways to redesign
their school cafeterias.

Text2: A new program lets students customize their meals,
participate in taste tests and brainstorm ways to redesign
their school cafeterias. Would you like to see this
program come to your school?

Jaccard similarity: 0.7933333333333333

Pair 1:

ID1: nyt://interactive/2731c978-c917-59c7-882f-2b533972a684

ID2: nyt://interactive/b990eefb-5633-536a-9077-76d005b5ab5b

Text1: Mr. Steyer interviews for The New York Times's endorsement.

Text2: Mr. Sanders interviews for The New York Times's endorsement.
Jaccard similarity: 0.7619047619047619

Pair 2:

ID1: nyt://interactive/2731c978-c917-59c7-882f-2b533972a684
ID2: nyt://interactive/93528953-1627-51c6-ac1f-5dcee23c97b7
Text1: Mr. Steyer interviews for The New York Times's endorsement.
Text2: Ms. Klobuchar interviews for The New York Times's endorsement.
Jaccard similarity: 0.7121212121212122

Pair 3:

ID1: nyt://interactive/2731c978-c917-59c7-882f-2b533972a684
ID2: nyt://interactive/8ce4ec22-cbe7-5c26-8603-9a7cf8c55b03
Text1: Mr. Steyer interviews for The New York Times's endorsement.
Text2: Mr. Patrick interviews for The New York Times' endorsement.
Jaccard similarity: 0.703125

Pair 4:

ID1: nyt://interactive/2731c978-c917-59c7-882f-2b533972a684
ID2: nyt://interactive/27bab2d4-745e-58c5-8a6d-cbeede182f11
Text1: Mr. Steyer interviews for The New York Times's endorsement.
Text2: Mr. Biden interviews for The New York Times's endorsement.
Jaccard similarity: 0.7580645161290323

3.1.4 Confusion results

True Positives (TP): 13

True Positives in Error Bound: 9

False Positives (FP): 5

=====

[False Positive #1]

ID 1 : nyt://article/2a8e3c7c-093e-5ae0-ba9d-b758e356f6a6
ID 2 : nyt://article/2801707a-4b94-50d2-84c3-69adf8e4c9cc
Jaccard Similarity : 0.4857 (Threshold: 0.7)
Text 1 : Ryan Patrick Smith makes his New York Times Crossword debut.
Text 2 : Barbara Lin makes her New York Times Crossword debut.

[False Positive #2]

ID 1 : nyt://article/2801707a-4b94-50d2-84c3-69adf8e4c9cc
ID 2 : nyt://article/6b9b6d12-2d49-5ed8-8257-5192b8411f8c
Jaccard Similarity : 0.4857 (Threshold: 0.7)
Text 1 : Barbara Lin makes her New York Times Crossword debut.
Text 2 : Michael Schlossberg makes his New York Times Crossword debut.

[False Positive #3]

ID 1 : nyt://article/aab4ed1e-3757-5b69-8e47-8961b2e5bf50
ID 2 : nyt://article/f1497094-54d3-5adf-a98a-6a0f3d8bc492

```

Jaccard Similarity : 0.5662 (Threshold: 0.7)
Text 1             : Splitting up, finding a car key and more reader
tales of New York City in this week's Metropolitan Diary.
Text 2             : Stargazing in Fort Tryon Park, leading a tour
group and more reader tales of New York City in this week's Metropolitan Diary.

[False Positive #4]
ID 1               : nyt://interactive/c891ed57-1184-52d9-bd9c-cf9f7bf23f13
ID 2               : nyt://interactive/b4967cb3-cbac-54d7-915f-d6d70ed3afa4
Jaccard Similarity : 0.5545 (Threshold: 0.7)
Text 1             : Here's how important Iowa is to the seven candidates
mounting competitive efforts there.
Text 2             : Here's how important New Hampshire is to the 10
Democrats mounting competitive efforts there.

[False Positive #5]
ID 1               : nyt://article/d9e74484-e06d-5d5a-9d90-dcde3e41c257
ID 2               : nyt://article/d1ea7ea3-1b7c-5a0c-9f46-257439f41094
Jaccard Similarity : 0.2740 (Threshold: 0.7)
Text 1             : If your preferred candidate doesn't win the nomination,
will you still do everything you can to deny Trump a second term?
Text 2             : Democrats need to nominate the right person to prevent
Trump from winning a second term.

```

3.2 Experiment 2

In this experiment we will test the pipeline using a 5-grams shingle based on chars. The desiderate threshold to 0.7 with an error of 0.1 so all solution with a Jaccard similarity ≥ 0.6 should be considered as true positive results.

3.2.1 Hyper parameters

```

K = 5
SHINGLER_TYPE = "char"
SHINGLER_BUCKET_SIZE_IN_BYTES=4
OPTIMIZATION_OBJECTIVE = "precision"
THRESHOLD = 0.7
ERROR = 0.1
MAX_SIGNATURE_LEN = 500

```

3.2.2 LSH parameters tuned on threshold

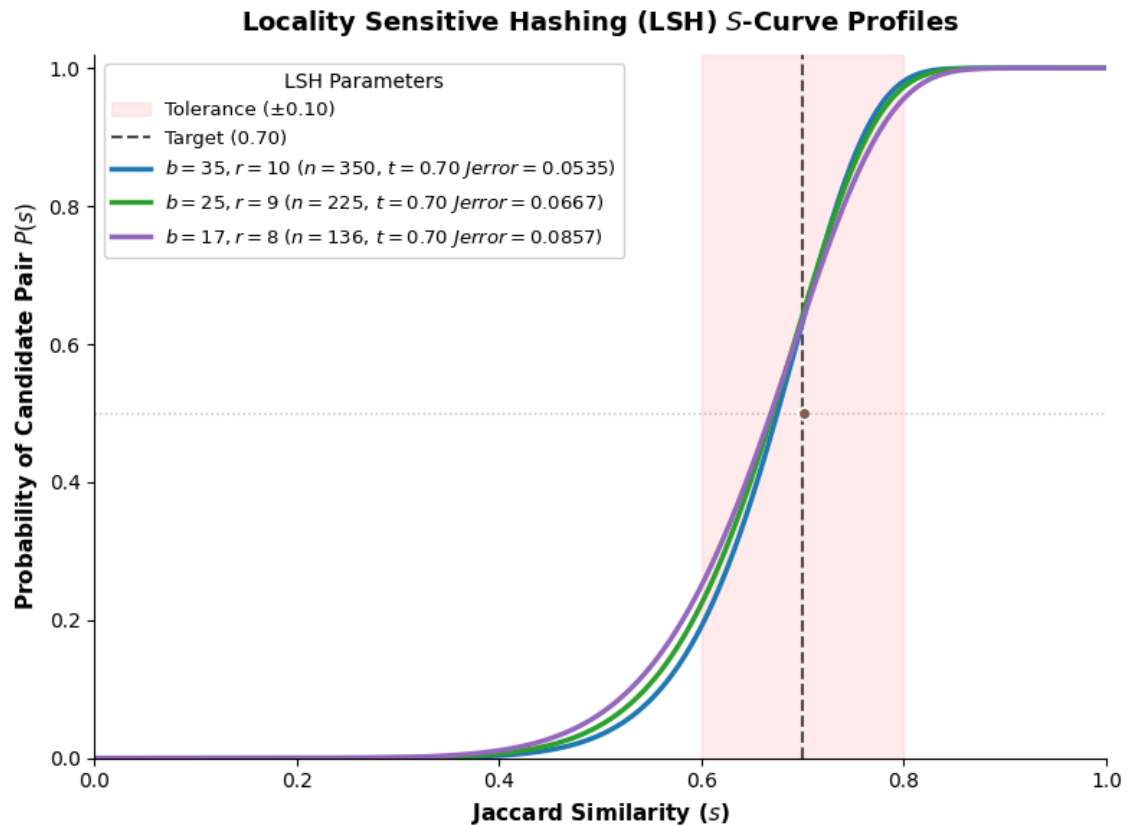
CLUSTER CONTEXT:

```

Shingles:
  k: 5
  type: char

```

hashed shingle bucket size in bytes: 4
Minhash:
signature length: 350
LSH:
bands: 35
rows: 10



3.2.3 Candidate pairs execution log

All candidate pairs found: 80

All non identical pairs found: 15

First 5 non identical pairs

Pair 0:

ID1: nyt://article/82928e0e-bc1f-5daf-8a62-027a45e486e1

ID2: nyt://article/547b64d3-7947-5205-a3a7-9dc2b7f31481

Text1: A new program lets students customize their meals,
participate in taste tests and brainstorm ways to redesign
their school cafeterias.

Text2: A new program lets students customize their meals,
participate in taste tests and brainstorm ways to redesign
their school cafeterias.

Would you like to see this program come to your school?

Jaccard similarity: 0.7352941176470589

Pair 1:

ID1: nyt://interactive/b990eefb-5633-536a-9077-76d005b5ab5b

ID2: nyt://interactive/064f1408-e21c-5e81-b575-97b3cc95498d

Text1: Mr. Sanders interviews for The New York Times's endorsement.

Text2: Mr. Yang interviews for the New York Times' endorsement

Jaccard similarity: 0.6349206349206349

Pair 2:

ID1: nyt://interactive/b990eefb-5633-536a-9077-76d005b5ab5b

ID2: nyt://interactive/8ce4ec22-cbe7-5c26-8603-9a7cf8c55b03

Text1: Mr. Sanders interviews for The New York Times's endorsement.

Text2: Mr. Patrick interviews for The New York Times' endorsement.

Jaccard similarity: 0.6060606060606061

Pair 3:

ID1: nyt://interactive/064f1408-e21c-5e81-b575-97b3cc95498d

ID2: nyt://interactive/8ce4ec22-cbe7-5c26-8603-9a7cf8c55b03

Text1: Mr. Yang interviews for the New York Times' endorsement

Text2: Mr. Patrick interviews for The New York Times' endorsement.

Jaccard similarity: 0.711864406779661

Pair 4:

ID1: nyt://interactive/b990eefb-5633-536a-9077-76d005b5ab5b

ID2: nyt://interactive/93528953-1627-51c6-ac1f-5dcee23c97b7

Text1: Mr. Sanders interviews for The New York Times's endorsement.

Text2: Ms. Klobuchar interviews for The New York Times's endorsement.

Jaccard similarity: 0.6666666666666666

3.2.4 Confusion results

True Positives (TP): 4

True Positives in Error Bound: 10

False Positives (FP): 1

=====

[False Positive #1]

ID 1	: nyt://interactive/93528953-1627-51c6-ac1f-5dcee23c97b7
ID 2	: nyt://interactive/8ce4ec22-cbe7-5c26-8603-9a7cf8c55b03
Jaccard Similarity	: 0.5882 (Threshold: 0.7)
Text 1	: Ms. Klobuchar interviews for The New York Times's endorsemen
Text 2	: Mr. Patrick interviews for The New York Times' endorsement.

3.3 Experiment 3

In this experiment we will test the pipeline using a 3-grams shingle based on words. The desiderate threshold to 0.7 with an error of 0.1 so all solution with a Jaccard similarity ≥ 0.6 should be considered as true positive results.

3.3.1 Hyper parameters

```
K = 3
SHINGLER_TYPE = "word"
SHINGLER_BUCKET_SIZE_IN_BYTES=8
OPTIMIZATION_OBJECTIVE = "precision"
THRESHOLD = 0.7
ERROR = 0.1
MAX_SIGNATURE_LEN = 500
```

3.3.2 LSH parameters tuned on threshold

CLUSTER CONTEXT:

Shingles:

k: 3

type: word

hashed shingle bucket size in bytes: 8

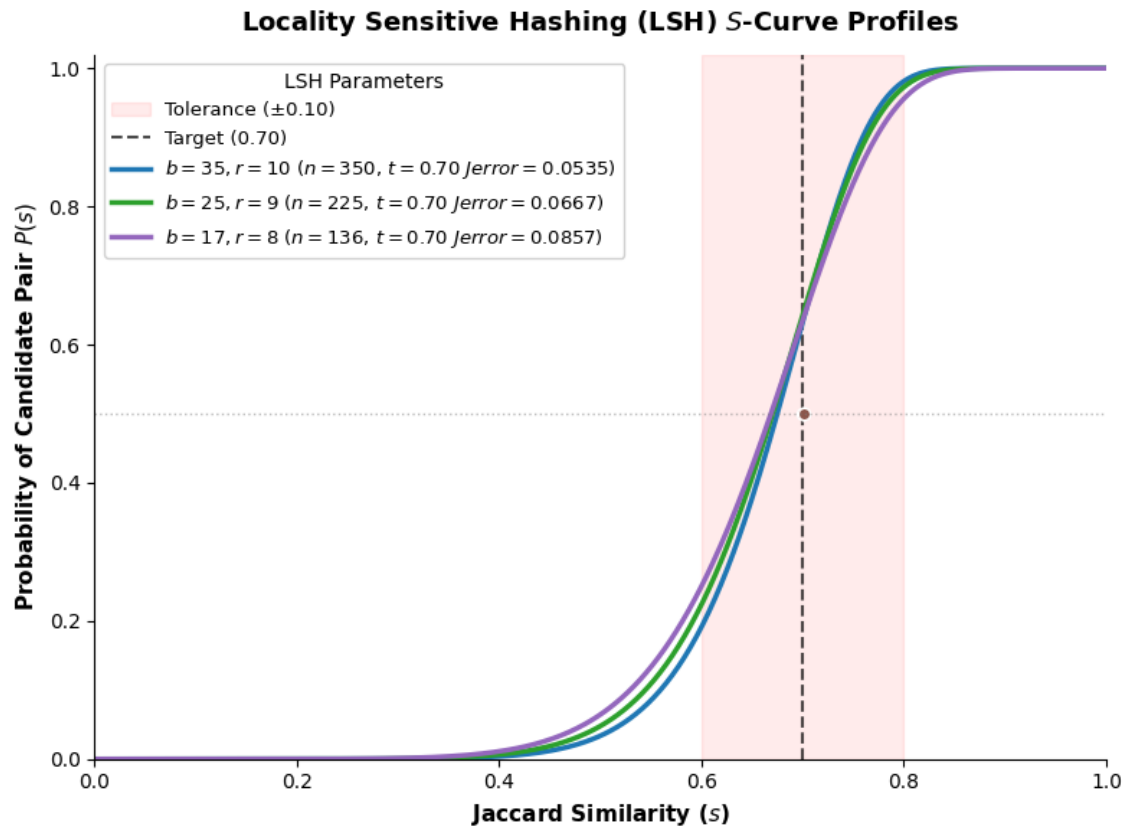
Minhash:

signature length: 350

LSH:

bands: 35

rows: 10



3.3.3 Candidate pairs execution log

All candidate pairs found: 73

All non identical pairs found: 8

First 5 non identical pairs

Pair 0:

ID1: nyt://article/82928e0e-bc1f-5daf-8a62-027a45e486e1

ID2: nyt://article/547b64d3-7947-5205-a3a7-9dc2b7f31481

Text1: A new program lets students customize their meals,
participate in taste tests and brainstorm ways to redesign
their school cafeterias.

Text2: A new program lets students customize their meals,
participate in taste tests and brainstorm ways to redesign
their school cafeterias. Would you like to see this program
come to your school?

Jaccard similarity: 0.6206896551724138

Pair 1:

ID1: nyt://interactive/b990eefb-5633-536a-9077-76d005b5ab5b

ID2: nyt://interactive/27bab2d4-745e-58c5-8a6d-cbeede182f11

Text1: Mr. Sanders interviews for The New York Times's endorsement.

Text2: Mr. Biden interviews for The New York Times's endorsement.

Jaccard similarity: 0.6

Pair 2:

ID1: nyt://interactive/b990eefb-5633-536a-9077-76d005b5ab5b

ID2: nyt://interactive/5c34d47f-f29c-5c22-b163-b811a3f6692e

Text1: Mr. Sanders interviews for The New York Times's endorsement.

Text2: Senator Warren interviews for The New York Times's endorsement.

Jaccard similarity: 0.6

Pair 3:

ID1: nyt://interactive/b990eefb-5633-536a-9077-76d005b5ab5b

ID2: nyt://interactive/d0598a7b-06ba-56db-9df4-7a2afa1c95e3

Text1: Mr. Sanders interviews for The New York Times's endorsement.

Text2: Mr. Buttigieg interviews for The New York Times's endorsement.

Jaccard similarity: 0.6

Pair 4:

ID1: nyt://interactive/5c34d47f-f29c-5c22-b163-b811a3f6692e

ID2: nyt://interactive/d0598a7b-06ba-56db-9df4-7a2afa1c95e3

Text1: Senator Warren interviews for The New York Times's endorsement.

Text2: Mr. Buttigieg interviews for The New York Times's endorsement.

Jaccard similarity: 0.6

3.3.4 Confusion results

True Positives (TP): 13

True Positives in Error Bound: 9

False Positives (FP): 5

=====

[False Positive #1]

ID 1 : nyt://article/2a8e3c7c-093e-5ae0-ba9d-b758e356f6a6
ID 2 : nyt://article/2801707a-4b94-50d2-84c3-69adf8e4c9cc
Jaccard Similarity : 0.4857 (Threshold: 0.7)
Text 1 : Ryan Patrick Smith makes his New York Times Crossword debut.
Text 2 : Barbara Lin makes her New York Times Crossword debut.

[False Positive #2]

ID 1 : nyt://article/2801707a-4b94-50d2-84c3-69adf8e4c9cc
ID 2 : nyt://article/6b9b6d12-2d49-5ed8-8257-5192b8411f8c
Jaccard Similarity : 0.4857 (Threshold: 0.7)
Text 1 : Barbara Lin makes her New York Times Crossword debut.
Text 2 : Michael Schlossberg makes his New York Times Crossword debut.

[False Positive #3]

ID 1 : nyt://article/aab4ed1e-3757-5b69-8e47-8961b2e5bf50
ID 2 : nyt://article/f1497094-54d3-5adf-a98a-6a0f3d8bc492
Jaccard Similarity : 0.5662 (Threshold: 0.7)
Text 1 : Splitting up, finding a car key and more reader
tales of New York City in this week's Metropolitan Diary.
Text 2 : Stargazing in Fort Tryon Park, leading a tour
group and more reader tales of New York City in this week's Metropolitan Diary.

[False Positive #4]

ID 1 : nyt://interactive/c891ed57-1184-52d9-bd9c-cf9f7bf23f13
ID 2 : nyt://interactive/b4967cb3-cbac-54d7-915f-d6d70ed3afa4
Jaccard Similarity : 0.5545 (Threshold: 0.7)
Text 1 : Here's how important Iowa is to the seven candidates mountin
Text 2 : Here's how important New Hampshire is to the 10
Democrats mounting competitive efforts there.

[False Positive #5]

ID 1 : nyt://article/d9e74484-e06d-5d5a-9d90-dcde3e41c257
ID 2 : nyt://article/d1ea7ea3-1b7c-5a0c-9f46-257439f41094
Jaccard Similarity : 0.2740 (Threshold: 0.7)
Text 1 : If your preferred candidate doesn't win the nomination,
will you still do everything you can to deny Trump a second term?
Text 2 : Democrats need to nominate the right person to prevent
Trump from winning a second term.

Bibliography

- [1] Jure Leskovec, Anand Rajaraman, and Jeff Ullman. *Mining of Massive Datasets*. Cambridge University Press, 2020.